S	M	T	W	T	F	S
MARCH 2015						
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Assignment Day 5 :- Registers & Counters.

(1) Design a counter with TFF whose count sequence 0, 2, 4, 5, 3, 1, 0 . . . . Since largest number is 5, so we need 3FF (Q_2, Q_1, Q_0)

present state	Next state
0 0 0	0 1 0
0 1 0	1 0 0
1 0 0	1 0 1
1 0 1	0 1 1
0 1 1	0 0 1
0 0 1	0 0 0

T-FF Excitation table :-

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$T = Q_n \oplus Q_{n+1}$

present state	Next state	T_2	T_1	T_0
0 0 0	0 1 0	0	1	0
0 1 0	1 0 0	1	1	0
1 0 0	1 0 1	0	0	0
1 0 1	0 1 1	1	1	0
0 1 1	0 0 1	0	1	1
0 0 1	0 0 0	0	0	1

$T_2 = Q_1 \bar{Q}_0 + Q_2 Q_0$

T_2

$Q_2 Q_1$ \ Q_0	0	1
0 0	0 0	0 4
0 1	1 2	0 6
1 1	x 3	x 7
1 0	0 1	1 5

$T_2 = Q_1 \bar{Q}_0 + Q_2 Q_0$

T_1

	0	1
00	1	0
01	1	1
11	x	x
10	0	1

$T_1 = Q_1 + \bar{Q}_0 + \bar{Q}_1 + Q_0 Q_2$

T_0

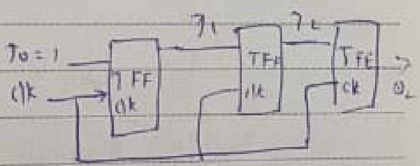
	0	1
00	0	1
01	0	0
11	x	x
10	1	0

$T_0 = Q_2 \bar{Q}_0 + \bar{Q}_2 \bar{Q}_1 Q_0$

(2) Design a 3 bit Up Down Counter using negedge triggered TFF

$M = 1 \rightarrow$ Up Count

$M = 0 \rightarrow$ Down Count



TFF Input eqn

LSB $T_0 = 1$

Middle $T_1 = M Q_0 + \bar{M} \bar{Q}_0$

MSB $T_2 = M Q_1 Q_0 + \bar{M} \bar{Q}_1 \bar{Q}_0$

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Thursday

(3) Circuit has 3-negedge trig T-FF
$T = 1$

o/p :- 1st FF : Q_0
2nd FF = Q_1
3rd FF = Q_2

Clk connections:-
FF0 is clocked by external clk.
FF1 is clocked by Q_0.
FF2 is clocked by Q_1

∴ Its 3-Bit Asynchronous (ripple) counter.

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Friday

clk pulse	Q_2	Q_1	Q_0	Dec	Count Sequence
Initial	0	0	0	0	0 → 1 → 2 → 3
1	0	0	1	1	6 ← 5 ← 4
2	0	1	0	2	7 → 0
3	0	1	1	3	
4	1	0	0	4	
5	1	0	1	5	
6	1	1	0	6	
7	1	1	1	7	
8	0	0	0	0	

3-Bit Asynchronous up (ripple) counter.